Preliminary System Design

DRAFT PENDING TEAM REVIEW after integration of all three received designs

Student Project Planning and Progress Tracking System

Prepared for team review by Wei Mingrui | Group9 Xingmang | ROUND2 | 25 October 2026

1 System design overview

This second-round draft integrates the actual 14-page F1/F2 report by Zhang Weiyi, the existing 11-page F3/F4 report by Wei Mingrui and the 31-page F5/F6 report by Zhang Yuyang. All source materials are retained. Substantive conflicts have been repaired in the three ROUND2 individual versions and shared contract; team/customer approval, detailed design and implementation remain pending.

The system supports small university coursework teams with shared projects, membership, task planning, progress records and understandable warnings. Core F1/F2/F5 are Must Have; F3 is Should Have; F4 and F6 are Could Have. Schedule estimates and risk overlays must not become prerequisites for basic task updates or progress viewing [S2, pp. 5, 19-21].

1 1 Evidence and source versions

Available evidence	Use and limitation
Chapter 6 lecture, 105 pages	Modular general-design principles. Treat course slides as reference material, not new task instructions.
Verified revised URS, 24 pages	Recovered from existing Git history and SHA-256 identical to the previously used attachment. Revised artifact confirmed; customer approval remains unproved.
Revised Task 3 analysis, 36 pages	Object-oriented analysis; identical bytes to the local REVISED PDF. Prior analysis work does not prove this week's design completion.
Proposal, 8 pages	Functional ownership and planning examples; later URS qualifications govern this draft.
Three actual individual designs	F1/F2 v1.0 14 pages; F3/F4 REVIEWED11 pages; F5/F6 v1.0 dated25 October31 pages. Compared and editorially integrated; original author approval not inferred.

1 2 General design boundary

This stage selects module responsibilities, dependencies, shared data and conceptual interfaces. Data structures, precise UI flows, storage schema, error names, transaction mechanisms and executable algorithms remain for detailed design. No implementation technologies are finalized and no application code is produced.

2 Overall module hierarchy

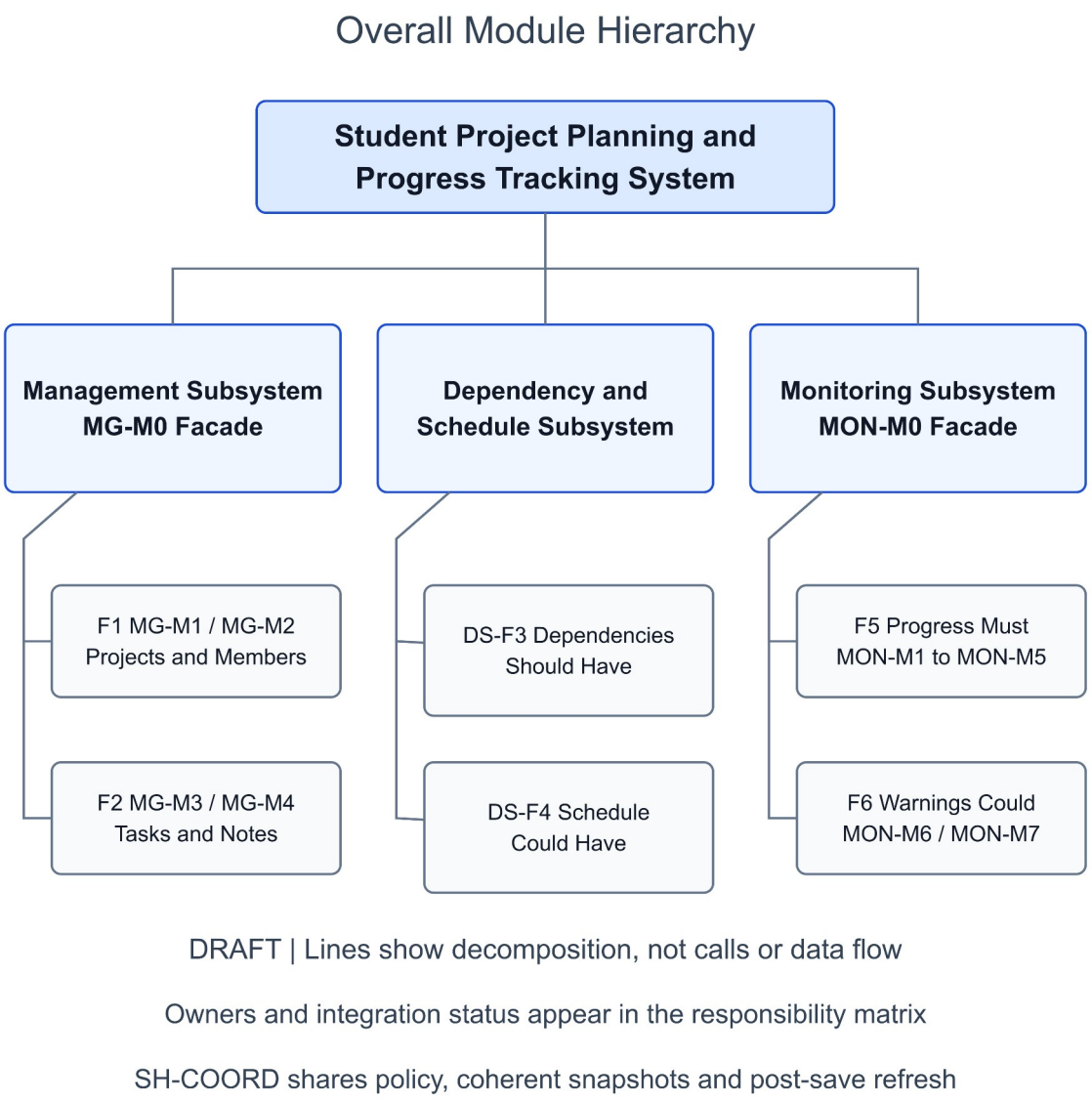


Figure1. Three software ownership areas with prefixed identifiers: MG-M0 facade and MG-M1-4; DS-F3/F4; MON-M0 facade and MON-M1-7. Lines are decomposition, not runtime flow. Names/material receipt/review status are recorded in the adjacent matrix; the design remains DRAFT.

User-facing screens, use-case coordination and persistence support all three areas. They are shared logical responsibilities in one proposed application, rather than extra functional ownership areas or independent services.

The preliminary design follows the use cases and domain objects from the Task 3 object-oriented analysis. There is no available DFD basis for a formal transform/transaction-derived Structure Chart. This diagram intentionally makes no such claim.

3 Responsibility and communication

3 1 Module responsibility matrix

Area and owner	Scope and interfaces	Current integration status
F1/F2 MG-M0-M4 Zhang Weiyi	Projects, membership/roles, task fields, assignment, status and notes. FR-01-FR-06. Provide coherent saved task/project data and permission decisions.	Source received; reconstructed editable ROUND2 from PDF; substantive shared-contract repairs applied; owner/team review pending.
F3/F4 DS-F3/F4 Wei Mingrui	Relationship validation and maintenance, cycle rejection, schedule calculation, critical activities and refresh. FR-07-FR-12, with FR-09 warning display in F5.	Existing editable source revised in ROUND2; original algorithms and17 scenarios preserved; protocol review pending.
F5/F6 MON-M0-M7 Zhang Yuyang	F5/F6 progress, assignments, blockers and warnings. FR-13 base -> MON-M0 / MON-M1 / MON-M2 / MON-M4 / MON-M5 + UI; Blocked subview -> MON-M3, dependent on Should FR-09/F3, D-01 pending. FR-14 completion; FR-15 optional three warning types.	Source received; reconstructed editable ROUND2 from PDF; priority/NFR/status/availability and UI repairs applied; owner/team review pending.
SH-COORD Shared facade responsibility	Consistent saves, snapshots, error mapping and review; all owners contribute to their end-to-end use cases.	One policy/snapshot/save/error protocol, realized by facades. Not an additional deployable service or a completed implementation.

3 2 Module communication and dependencies

A user action enters the use-case coordinator, which checks project membership and operation permission before detailed task validation. Unknown or inaccessible references share the user-visible message "Invalid or inaccessible task." Do not disclose private names, titles, membership or existence. F1/F2 provide task/project data; F3 validates the candidate. Valid source changes save atomically before derived views refresh. Explicit calls are sufficient; no event broker is required.

F4 reads trustworthy F2 durations and F3 links, and alone calculates CPM/finish/critical facts. F5 reads task statuses plus known F3 links, derives direct Blocked reasons and basic progress/overdue facts. F6 composes only FR-15 warnings from F5 facts and current available F4 results. No monitoring module writes source links, statuses or assignments.

Cross-owner FR-09 contract: F3 owns dependency validation and saved prerequisite relationships. F5 combines those valid links with F1/F2 task-status records to derive Blocked warnings and named incomplete prerequisites. F5 must not modify F3 records. Dependency or status changes request a F5 refresh. Basic F3 and F5 functions work without F4. The one-way links/status-to-warning flow avoids circular module dependencies; F4/F6 consume source data or derived facts without writing them.

4 Modularity and information hiding

4 1 High cohesion and low coupling review

Design concern	Proposed treatment and review
Functional cohesion	F3 validates and maintains one relationship graph. F4 analyzes one valid plan. F1/F2 own editable records; F5/F6 own interpretation and presentation of progress/risk.
Data coupling	Exchange explicit project/task IDs and read-only snapshot/result records. Do not share mutable global arrays, UI controls or storage entities across domains.
Control coupling	Coordinator requests a named operation after a valid save. Modules do not pass broad mode flags that change unrelated behaviors.
Optional F4/F6	F3 and F5 remain usable when schedule analysis is deferred. F6 must mark critical/deadline-estimate warnings unavailable when the needed F4 result is absent.
Small-team cost	Implement as ordinary internal modules in one application. Separate private graph helpers only where they avoid duplication. Do not equate every leaf with a service or class.

4 2 Information hiding strategy

Task Management hides record validation, membership storage and task lifecycle details. Dependency Management hides adjacency representation, traversal markers and relationship persistence. Schedule Analysis hides topological tie-breaking, timing arrays and criticality calculations behind a result contract. Monitoring hides display aggregation and warning composition.

Shared application coordination hides save ordering and revision checks. A storage adapter hides database/file layout. The UI receives readable outcomes and availability states; it does not duplicate domain calculations. No module may directly alter another owner's hidden records through an exposed mutable object.

4 3 Chapter 6 application

Chapter 6 distinguishes general from detailed design and discusses modularization, abstraction, information hiding, independence, cohesion and coupling [S1; page-specific references in the source register]. These principles apply to this project even though Task 3 used object-oriented requirements analysis. Functional ownership and the use-case/domain model provide the decomposition basis.

The next review will test whether each interface carries only the data needed and whether a local change can remain inside one owner's module. This review is planned; the teammate boundaries shown here remain proposals until accepted.

5 Preliminary data and interface overview

Conceptual record	Minimum information and owner
Project and Membership	ProjectSnapshot: projectId/name/plannedStartDate/projectDeadline/members/tasks/dependencyView/revision. MembershipRecord distinguishes memberId from userId; permissions project-specific. F1/F2 source ownership.
Task and ProgressNote	TaskSnapshot taskId/projectId/name/assigneeId/durationDays/taskDeadline/status; TaskStatus NOT_STARTED|IN_PROGRESS|COMPLETED. ProgressNote noteId/taskId/authorMemberId/text; retention/editing policy open. F2 owns source.
Dependency	DependencyResultView: availability/sourceRevision/edges/reasonCode; sourceRevision equals coherent snapshot.revision. F3 owns same-project links, no status writes. Reason scope and proof-based current publication are in Sections5.2/11; CALCULATION_FAILED belongs to schedule analysis, not dependency reads.
ScheduleResultView	ScheduleResultView: availability/sourceRevision; nullable duration/timings/criticalTaskIds/estimatedProjectFinish. NOT_ENABLED uses FEATURE_NOT_ENABLED; UNAVAILABLE uses INVALID_INPUT, CALCULATION_FAILED, STALE_RESULT or DEPENDENCY_INPUT_UNAVAILABLE. VALID/NO_TASKS have no failure reasonCode. F4 alone calculates; finishReasonCode is separate. Authorized INVALID_INPUT can carry DurationInputDiagnostic[]; timing/critical/finish facts stay null. Diagnostics=null for other states; they never bypass generic access errors.
Progress and Warning views	DashboardSummary has progress/rawStatusCounts, separate BlockedResult, assigned/overdue views, schedule and warnings with per-type evaluations. F5 derives blockers/counts; F6 only composes retained warning types.

5 1 Shared boundary contracts

F1/F2 -> consumers: one authorized ProjectSnapshot with canonical TaskSnapshot/MembershipRecord data, projectDeadline/taskDeadline/plannedStartDate and revision. SH-COORD combines coherent F3 dependency reads; existing-project writes require expectedRevision. Implementation transport/schema remain provisional.

F3 -> F5/F4: DependencyResultView with availability/sourceRevision/edges; F4 -> monitoring: ScheduleResultView with current sourceRevision, nullable payload and explicit reasons. F5 -> F6: current rawStatus, blocked direct-prerequisite facts and basic overdue results. F4/F6 feature-off never disables core source/status/completion views.

MG-M0 exposes the original seven management operations plus an explicit deleteTask lifecycle contract already present in F2 scope. DS-F3/F4 retain seven graph/schedule operations. MON-M0 coordinates six monitoring query contracts from the received report. Shared names and result envelopes are now compared and defined; these are conceptual operations, not APIs or owner-approved implementation decisions.

5 2 Data consistency and failure behavior

Every successful atomic source save advances revision once; failures preserve input and prior source. SH-COORD binds projectId, tasks, memberships and complete F3 edges to one coherent revision. DependencyResultView.sourceRevision must equal ProjectSnapshot.revision. After unrelated notes/roles change, reauthorize and prove task universe/validated edges unchanged, or re-read coherently, before republishing at the current sourceRevision. Relevant task/link changes require graph validation. Failed proof -> dependency UNAVAILABLE/STALE_RESULT with edges=null. Schedule reuse separately requires unchanged task IDs/graph/durations and remapped plannedStartDate; otherwise schedule UNAVAILABLE/STALE_RESULT. No second counter or implemented transaction is claimed; protected version/cycle/save constraints remain design requirements.

6 Major decisions and unresolved questions

6 1 Major design decisions

Preserve three functional owner areas. Use a directed acyclic task graph and full O(V + E) calculation for retained F4. Reject cycle-producing candidate edits before commit. Keep Blocked separate from the stored task status. Use calendar days, optional parallel work and human workload decisions. Keep F4/F6 optional and all technology choices provisional.

Source reconciliation: the pre-review Group9_Scaffold/docs/ARCHITECTURE.md Overview row said "four-category counts" and "each task only one display category"; connected monitoring documentation explicitly listed Blocked as a mutually exclusive display category. This conflicts with the revised URS warning-overlay model. A small documentation-only correction now uses stored-status counts plus an independent Blocked warning, retains manual status updates and counts each task once for completion. Backups and exact affected wording are recorded in the design review. No application behavior or stored status was changed.

6 2 Open technical and requirement decisions

Issue	Decision and next owner
Baseline and scope	Revised URS artifact verified; formal priorities unchanged. Team/customer decide FR-13 blocker dependency on Should F3 and retained F4/F6 Could scope; file identity is not approval.
Permissions	Zhang Weiyi + Wei: dependency-edit rights, invitations and leader-agreement mechanism (OQ-06). No assumed approval workflow.
Task validity and lifecycle	Zhang Weiyi + Wei: required fields, duration positivity/precision, deletion/link cleanup and retained assignments (OQ-05).
Status and exceptions	All owners + customer: status vocabulary, early completion, real-work early starts versus finish-to-start calculation, empty schedule and empty progress (OQ-08).
Date comparisons	Wei + Zhang Yuyang: exact finish/deadline inclusion, date precision and timezone for overdue checks. Deadline is a target only.
Performance and recovery	All owners: capacity, concurrency, devices/network/load and measurement of 1-2 second expectation; save/recovery criteria (OQ-02/OQ-07).
Technical design	All owners: language/framework/database/deployment, save transaction/revision mechanism and detailed interface format.
Other URS open scope	Customer/team: export, undo, mobile acceptance, hours/sample/soft hints (OQ-01/OQ-03/OQ-04/OQ-09). Do not assume these features approved.

These remain decisions to obtain, not approvals. In particular FR-13 Must blocker visibility depends on Should FR-09/F3; the fallback is truthful partial capability, not a formal resolution or priority upgrade. Duration/field policy, lifecycle, early completion, dates, capacity, recovery and high-priority assignment wording remain open.

7 Received design inputs and remaining review

7 1 Source receipt and integration checklist

Owner	Requested input and acceptance check
Zhang Weiyi F1/F2	Received14-page source; preserved in Sources. ROUND2 retains project/member/task/note modules, repairs F3/F5 ownership, facade hierarchy,8 write contracts, identity/date/revision/error boundaries and lifecycle references. Await owner/team review.
Zhang Yuyang F5/F6	Received31-page source; preserved in Sources. ROUND2 retains13sections and10 scenarios, repairs formal priorities/NFR-05, raw counts plus Blocked, partial dashboard, version/availability and both figures. Await owner/team review.
Wei Mingrui F3/F4 and integration	F3/F4 source preserved and revised; shared-contract/27-scenario integration and numerical checks prepared. Remaining customer/team decisions are explicit; no approval or software-test claim.

7 2 Next stage design plan

1. Review the received source designs and ROUND2 editorial changes with each functional owner. Receipt and static integration are complete; agreement and customer approval are not inferred.

2. Confirm the shared dictionary, permissions, revision/availability protocol, Blocked overlay, dates and optional scope, including the FR-13 versus FR-09/F3 dependency decision.

3. Walk through the27 NOT RUN integration scenarios and verify source-to-revision mappings. The three source designs are integrated for review; implement nothing until the agreed next stage.

4. Produce detailed module/interface/data/UI design and agreed test oracles. Preserve the 8-day, 10-day, equal-critical-path, failure and resource-boundary cases.

5. Only after scope and design agreement, plan implementation and software testing. This week's package does not claim those activities. Next-week target period is 26 October to 1 November 2026; task allocation and completion dates require team agreement.

7 3 Integration exit criteria

Retain DRAFT/PENDING REVIEW until the team actually accepts the module boundaries/contracts and customers resolve required scope/assumption decisions. All three source documents are received; no signatures, review meeting, acceptance or test outcome is invented.

8 Source register and review record

[S1] SE_Chapter6-Structured Software Design-2026.10.22.pdf, 105 PDF pages. Stored at 00_Source_Materials/Course_Lectures/SE_Chapter6-Structured Software Design-2026.10.22.pdf. Page references use physical PDF pages.

[S2] Student_Project_Planning_and_Progress_Tracking_System_URS_REVISED.pdf, 24 pages, dated 10 October 2026. Recovered from Requirements_Specification Git commit 148981bced19f2dc2e16ca4d326e2fc1e5f78b60, path outputs/Student_Project_Planning_and_Progress_Tracking_System_URS_REVISED.pdf, into 00_Source_Materials/Task2_URS/. Its SHA-256 matches the URS(1).pdf used in the original design. The revised artifact is verified; customer approval is not inferred. Relevant pp. 12-14, 18-21, 24.

[S3] Student_Project_Planning_and_Progress_Tracking_System_Requirements_Analysis_REVISED.pdf, 36 pages, dated 10 October 2026. Available at Requirements_Analysis/outputs/. Its SHA-256 matches the previously supplied Requirements_Analysis(1).pdf; see the review evidence manifest. Relevant PDF pp. 7, 19-21, 26-29, 33-35.

[S4] Student_Project_Planning_and_Progress_Tracking_System_Proposal(1).pdf, 8 pages. Stored at 00_Source_Materials/Task1_Proposal/Student_Project_Planning_and_Progress_Tracking_System_Proposal(1).pdf. The separately requested Proposal_FINAL filename was not found. Relevant pp. 3-5 planning rules and worked example; p. 8 functional ownership.

[S5] Current Group9_Scaffold documentation and dated22 October review record. Pre-round Git HEAD is 10e23fa929e23aa2816f370f4bd1be88d4171831; ROUND2 documentation edits are uncommitted. Historic scaffold notes are background, not authority over URS.

8 1 Lecture evidence

[S1] PDF pp. 8-10 distinguish general design from detailed design and state the general-design scope; pp. 23-25 explain internal cohesion and intermodule coupling; p. 29 describes information hiding; p. 37 recommends focused modules, simple interfaces and encapsulation. PDF pp. 40-43 present hierarchy as decomposition, while pp. 45-46 add calling and message details for a Structure Chart. Transformation starts from a refined DFD (pp. 49, 51); transaction analysis requires a transaction center (p. 67). We apply the modular principles and do not claim either unavailable flow-analysis basis.

8 2 Current review record

Updated25 October 2026 after the22 October review. Actual source designs, requirement clauses, interfaces, pseudocode, wireframes and sample arithmetic were compared. ROUND2 design files and five attachments record the repaired editorial protocol; document rendering/static checks do not validate application behavior.

Pending: actual functional-owner/team review, customer approval and scope decisions, detailed design, implementation and software-test execution. Every integration scenario is NOT RUN. Historical22 October records remain unchanged; the current25 October weekly snapshot records newly received materials and completed document work.

[S6] Zhang Weiyi, 01_Project_Task_Module_Design(1).pdf, 14 pages, Version 1.0, Week2026-10-22. Original retained in Individual_Designs/ZhangWeiyi/Sources/. Received and compared for this round; source has no corresponding editable DOCX here.

[S7] Zhang Yuyang, 03_Progress_Risk_Module_Design(1).pdf, 31 pages, Version 1.0,25 October 2026. Original retained in Individual_Designs/ZhangYuyang/Sources/. Received and compared for this round; editorial corrections do not imply owner approval.

9 Canonical shared dictionary and interfaces

Shared record	Fields and source ownership
ProjectSnapshot	projectId, name, plannedStartDate|null, projectDeadline|null, members, tasks, dependencyView, revision. SH-COORD obtains the coherent view; F1/F2 own base records and F3 owns relationships.
TaskSnapshot	taskId, projectId, name, assigneeId:MemberId|null, durationDays, taskDeadline|null, status:TaskStatus. Mandatory fields and duration precision remain open, separate from F4 analysis preconditions. Recommended exact decimal/canonical precision adds no integer-only business rule; F2 save policy stays separate.
Identity and notes	actorId/viewerId:UserId; memberId is a project membership ID. Resolve viewerMemberId before matching assigneeId. Preserve historical note authorMemberId when active membership is removed.
TaskDisplayView	rawStatus plus independent nullable isBlocked/isCritical and known blocker reasons. No four-way DisplayCategory or stored BLOCKED state.
ProgressResult	availability VALID|NO_TASKS, sourceRevision, totalTasks/completedTasks, percentage|null. Three raw counts partition tasks; blocked overlay never changes completion denominator.
Public results	MutationResult<T> succeeds with revision/SavedChange or fails with PublicError. DashboardResult succeeds with DashboardSummary or fails only for access/unsafe source failure; optional analysis problems are availability states.

Canonical field usage replaces legacy submissionDeadline/project.deadline with projectDeadline, Task.deadline with taskDeadline, plannedStart with plannedStartDate and calculatedProjectFinish with estimatedProjectFinish. These aliases appear only in change/evidence mappings, never as rival active fields. revision/expectedRevision/sourceRevision reference one project-local counter.

Boundary	Conceptual operation families and ownership
Management	MG-M0/M1-4: createProject/updateProject/manageMembership/createTask/updateTask/ updateTaskStatus/addProgressNote/deleteTask. Every existing-project write has expectedRevision; policy belongs to MG-M2.
Dependencies	DS-F3: addDependency/removeDependency/validateDependency/detectCycle. Valid saved edges supplied to F4/F5; candidate validation is not a current schedule publication.
Schedule	DS-F4: calculateSchedule/identifyCriticalActivities/recalculateSchedule. Produces ScheduleResultView; monitoring consumes it instead of recalculating CPM.
Monitoring	MON-M0..7: calculateProgress/identifyBlockedTasks/getAssignedTasks/detectOverdueTasks/ getScheduleWarnings/buildDashboardSummary. Query results carry sourceRevision/availability; no source writes.

10 Version consistency and normal save flow

One proposed project revision protocol applies across all three areas. SH-COORD is a shared responsibility realized by MG-M0/MON-M0, not an extra service. MG-M2 supplies the common project membership/edit policy.

Stage	Design invariant
Read	Authenticate/authorize the project context, then read one coherent ProjectSnapshot of F1/F2 records and F3 links at revision r. Do not merge independent current reads without consistency proof.
Mutate	Every existing-project write supplies expectedRevision=r. The protected commit checks authorization, current revision and all invariants; dependency cycle validation belongs to that same decision.
Commit	Create project at revision 1. Each successful atomic source change advances the revision once; a batch such as member departure/unassignment is one change. Rejected/failed requests do not increment or publish success.
Refresh	SavedChange names projectId, revision, changeKind, affectedTaskIds and changedFields. Use ordinary coordinator calls after success, without introducing a broker.
Derive	Use sourceRevision to reference that same counter. F5/F6 only use facts matching the current snapshot. sourceRevision is not an independent version field.
Reuse	Prove task IDs, graph and durations unchanged before reusing relative timings. Remap estimatedProjectFinish if plannedStartDate changes; refresh labels/comparisons and republish at current sourceRevision. Otherwise mark STALE_RESULT unavailable.

Concurrent reverse-edge requests from revision r cannot both commit: one succeeds or conflicts; the next attempt must read and revalidate against the resulting graph. Storage/locking technology and measurable performance remain detailed-design decisions. A change of asOfDate is a read and does not advance revision.

11 Failure flows and capability states

Availability	Meaning and payload
VALID	Nonempty valid schedule at current sourceRevision: durationDays, timings and complete criticalTaskIds. estimatedProjectFinish is null with a finishReasonCode when absolute date inputs/mapping are unavailable; relative timings remain usable.
NO_TASKS	Enabled analysis with no tasks. Schedule durationDays/criticalTaskIds/estimatedProjectFinish are null. Progress totalTasks=0, completedTasks=0, percentage=null. Do not fabricate a delivery estimate or0%.
NOT_ENABLED	The optional feature is not retained/enabled. Payload is null; reasonCode=FEATURE_NOT_ENABLED. Distinct from a known empty result.
UNAVAILABLE	Use reasonCode INVALID_INPUT, CALCULATION_FAILED, STALE_RESULT or DEPENDENCY_INPUT_UNAVAILABLE. Current schedule facts are null; no stale highlights or finish warnings are presented as current. Only authorized INVALID_INPUT may include task/field diagnostics, not partial schedule facts.

DependencyResultView: F3 off -> NOT_ENABLED/FEATURE_NOT_ENABLED, edges=null; enabled empty tasks -> NO_TASKS, edges=[]; enabled nonempty complete valid current input -> VALID, including edges=[] for no prerequisites. UNAVAILABLE reasons: DEPENDENCY_INPUT_UNAVAILABLE (missing complete read), INVALID_INPUT (invalid graph), STALE_RESULT (coherence unproved); not CALCULATION_FAILED. F4 gates: off -> NOT_ENABLED; enabled empty -> NO_TASKS; enabled nonempty without trustworthy same-revision dependencies -> UNAVAILABLE/DEPENDENCY_INPUT_UNAVAILABLE. edges=null is never an empty graph.

Schedule-only failure preserves base progress, raw statuses, assignments and MON-M5 overdue facts. With current safe tasks and valid taskDeadline/asOfDate, enabled F6 can evaluate OVERDUE_TASK independently. BLOCKED_CRITICAL_TASK and PROJECT_DEADLINE_CONFLICT stay unevaluated when required current critical/finish inputs are absent. Unknown is not no risk; evaluationByType states the limitation. F6 off leaves FR-15 NOT_ENABLED, while basic FR-13 overdue facts remain available. Only unsafe/inaccessible source data can fail the whole dashboard.

Only unavailable/unsafe source reads or inaccessible project contexts justify a top-level failure. Unknown/inaccessible project/task references receive the generic public messages; detailed authorized graph errors remain useful. NFR-05 is Reliability: preserve entered input and valid saved records, reject invalid edits without replacing the valid plan, and plan recoverability after temporary unavailability.

11 1 Numeric precision and authorized diagnostics

Proposed NumericPolicy under PENDING D-03/D-07 preserves canonical positive finite decimal inputs and all computation precision; no intermediate or display rounding feeds CPM. Preferred exact decimal/rational arithmetic certifies mathematical totalFloat=LS-ES=0. A binary backend must use one centralized, validated error-interval policy from conversions/operations/range/graph propagation. A near-zero interval requires canonical exact or higher-precision proof, not an arbitrary epsilon or automatic criticality. Any genuine positive float remains noncritical even if displayed0.00. Unresolved reliable timing/criticality yields whole UNAVAILABLE/CALCULATION_FAILED, never partial current critical IDs. F5/F6 consume only F4-certified facts.

For analysis, every duration must be present, numeric, finite and positive. Invalid durations yield ScheduleResultView UNAVAILABLE/INVALID_INPUT with null duration/timings/critical/finish and optional authorized diagnostics {taskId, field:durationDays, issue:MISSING|NON_NUMERIC|NON_FINITE|NON_POSITIVE, message}. Other states have diagnostics=null. Detailed task diagnostics require an accessible coherent project context; unknown/denied references use generic PublicError with no private payload. Missing-duration core-save policy remains D-03; F4 failure does not itself reject or roll back a valid F2 save. Reliable F5 counts, assignments, overdue facts and current F3/status-based MON-M3 blockers continue; only unsupported F6 types are unevaluated.

11 2 Conditional date mapping

D-04 remains PENDING. Recommend calendar days, Day0=plannedStartDate and estimatedProjectFinish=plannedStartDate+ceil(T) from canonical unrounded NumericPolicy-resolved T. Never ceil a display value or unresolved binary excess above an integer. Known unfinished task overdue uses asOfDate>taskDeadline; known current project conflict uses estimatedProjectFinish>projectDeadline; equality triggers neither. Missing planned start uses MISSING_PLANNED_START; absent/unconfirmed mapping uses DATE_MAPPING_UNRESOLVED and null finish. Other missing dates leave comparisons unevaluated. Project calendar/timezone and deadline inclusion still require agreement. IT12/16/21/22 explicitly select the candidate only conditionally; no client/chat timezone or approval is invented.

12 Formal priorities and dependency decisions

Requirement	Formal priority	Input dependency and honest fallback
FR-09 Blocked	Should Have	F2 statuses plus valid F3 direct edges. F3 off/unavailable -> unknown blocker view; do not upgrade priority.
FR-10 Duration	Could Have	Current finite positive durations and a complete valid current graph. F4 off -> NOT_ENABLED; enabled empty tasks -> NO_TASKS; nonempty unknown dependency input -> UNAVAILABLE/DEPENDENCY_INPUT_UNAVAILABLE; invalid schedule input -> UNAVAILABLE/INVALID_INPUT.
FR-11 Critical activities	Could Have	Current valid F4 result. Null when unevaluated, not false/empty known set.
FR-12 Refresh	Could Have	Retained F4 refresh after successful relevant saves; deadlines update comparisons and status refreshes source-based views.
FR-13 Dashboard	Must Have	Base -> MON-M0 / MON-M1 / MON-M2 / MON-M4 / MON-M5 + UI. Blocked subview -> MON-M3; depends on Should FR-09/F3. F3 off/unavailable gives unknown blockers, not none. D-01 remains pending; base fallback does not imply full FR-13 acceptance.
FR-14 Completion	Must Have	Saved three-state values; completed/total*100; empty percentage null; overlay not counted twice.
FR-15 Warnings	Could Have	Optional layer with3 supported types and per-type availability. FR-13 basic overdue facts do not make FR-15 Must.

Exact non-functional names: NFR-01 Usability; NFR-02 Browser and Device Support; NFR-03 Performance; NFR-04 Security and Access Control; NFR-05 Reliability. Maintainability/testability/explainability are design quality goals without a new URS ID. NFR-05 covers visible save failure, retained input, preserved valid records/plan and recoverability. Candidate date strict> comparisons, empty wording, field precision, lifecycle/leader policy, high-priority assignment meaning and measurable acceptance remain open.

Normal path: authorized request -> coherent snapshot -> expectedRevision/invariant validation -> atomic source commit -> revision/SavedChange -> F4/Blocker/progress refresh -> sourceRevision-gated dashboard. Failure paths preserve prior source on rejection/save failure; a valid save followed by F4 failure keeps new source and returns a partially capable dashboard. Access or unsafe source failure returns no private overview. No application behavior is claimed implemented.

13 Cross module validation matrix

All27 scenarios below are NOT RUN design plans. The independent Integration_Test_Matrix_ROUND2.md records each precondition, input, action, expected result, modules and requirement trace. These summaries do not report test outcomes.

ID	Scenario	Expected design outcome	Trace
IT01	Create project and task facts	Project revision starts at1; successful atomic task save advances it once. TaskSnapshot has projectId/taskId, canonical fields and proposed NOT_STARTED status; no BLOCKED field is persisted. Missing-duration task-save behavior follows agreed F2 policy, not F4 enablement.	FR-01/03/06; NFR-04/05
IT02	Accept valid dependency and refresh	F3 owns the saved edge; revision becomes r+1 once; SavedChange appears only after success. F5 refreshes from current statuses/links. Retained F4 recalculates or validly republishes; it never derives Blocked.	FR-07/09/12; NFR-05
IT03	Reject cycle without replacing valid plan	Reject A->B->C->A with an authorized explanation. Existing links, revision and valid prior schedule remain unchanged; no successful SavedChange or partial write.	FR-08/12; NFR-05
IT04	Blocked is an independent warning	B isBlocked=true and names A plus A's raw status. B's stored status is unchanged. No transitive prerequisite is invented; COMPLETED dependents are absent from unfinished blocked work.	FR-04/09; DR-04
IT05	Completing a prerequisite clears its reason	C disappears from D's incomplete direct prerequisite list; D isBlocked=false when no incomplete direct prerequisites remain. Link records and entered durations remain unchanged.	FR-04/09/12; DR-04
IT06	Status refresh does not shorten duration	Revision advances on accepted save; progress, assignment and blocker views refresh. Durations, links and resource assignments are not inferred or changed. Unsupported early-completion permission is not assumed.	FR-04/09/13/14; DR-01/04/05
IT07	Two of four tasks means fifty percent	Completed2, IN_PROGRESS1, NOT_STARTED1, total4, percentage50. Blocked overlay count1 does not subtract a raw count or add a fifth denominator entry.	FR-09/13/14; DR-04
IT08	Empty project is not zero-percent completion	With F4 enabled and tasks=[], progress is NO_TASKS with completed=0/total=0/percentage=null. F4 returns NO_TASKS before the dependency-input gate, with null duration/critical IDs/finish and no failure reasonCode. No division by zero or misleading delivery guarantee.	FR-10/11/13/14; OQ-08

13 Cross module validation matrix continued 9 to 16

ID	Scenario	Expected design outcome	Trace
IT09	F4 off keeps basic dashboard usable	F4 off -> NOT_ENABLED/FEATURE_NOT_ENABLED with null schedule payload; not dependency failure. Base progress, statuses, assignments, known blockers and MON-M5 overdue facts remain. isCritical=null; unsupported critical/finish warnings are unevaluated. F6 off also gives NOT_ENABLED; basic FR-13 overdue does not depend on FR-15.	FR-09/13/14; FR-10/11/12/15 Could scope
IT10	Unavailable or failed analysis is partial degradation	Safe current tasks and valid taskDeadline/asOfDate: F4 UNAVAILABLE preserves base status, assignment, progress and MON-M5 overdue facts. Enabled F6 can evaluate OVERDUE_TASK independently; absent current critical/finish inputs make BLOCKED_CRITICAL_TASK and PROJECT_DEADLINE_CONFLICT unevaluated. Unknown is not no risk. F6 off keeps FR-15 unavailable. Current valid F3/status inputs also keep MON-M3 Blocked facts and direct incomplete-prerequisite reasons available; if F3 validity cannot be proved, use IT18.	FR-10/11/13/15; NFR-05
IT11	Reject stale schedule facts	With nonempty safe coherent tasks, F4 enabled and complete current dependencies, a prior schedule at r cannot be used at r+1. Prove relevant schedule inputs still valid and republish current results, or return UNAVAILABLE/STALE_RESULT with null payload. Keep the base dashboard; dependency failure is separately IT18. Valid current F3/status inputs retain MON-M3 Blocked facts and direct prerequisite reasons despite stale/unavailable F4. Unproven dependencies instead give UNAVAILABLE/null blockers under IT18; never reuse stale Blocked data.	FR-12/13/15; NFR-05
IT12	Project-level deadline conflict uses valid finish	Conditional on the selected D-04 calendar/ceil(T) candidate: Base no conflict; changed plan one-day project conflict. Do not compare this project warning against a taskDeadline or recalculate CPM in MON-M7. Equality is a separate candidate boundary case.	FR-10/12/15; DR-02; date decision D-04
IT13	Private project and task references do not leak	No private payload/names/titles/membership/existence. Unknown and inaccessible projects share Invalid or inaccessible project.; tasks share Invalid or inaccessible task. Accessible members without edit rights receive the agreed permission result without mutation.	FR-06; NFR-04
IT14	Failed save preserves input and valid source	SAVE_FAILED; caller retains entered input; prior valid records/revision/plan remain; no success fact. A later derived-analysis failure after a successful source save is IT10, not a rollback of an already valid source commit.	NFR-05; FR-08/12
IT15	Concurrent dependency writes cannot jointly create cycle	At most one commits without rereading. Other request conflicts or revalidates against the new graph and is rejected as a cycle. No mixed candidate or increment on rejected request.	FR-08/12; NFR-05
IT16	Deadline changes refresh the correct facts	Conditional on the selected D-04 calendar/ceil(T) candidate: Task date refreshes MON-M5 basic overdue and optional FR-15 overdue warnings; project date refreshes MON-M7. Neither deadline constrains CPM or changes durations. Republish reused schedule only after current-input proof.	FR-01/03/12/13/15; date decision D-04

13 Cross module validation matrix continued 17 to 24

ID	Scenario	Expected design outcome	Trace
IT17	Formal priorities are preserved	FR-09 Should; FR-10/11/12 Could; FR-13/14 Must; FR-15 Could. Distinguish basic overdue facts from optional warning composition. FR-13 blocker dependency on Should F3 is recorded, not silently resolved by changing priorities.	URS pp.7-8/15-16/20; decision D-01
IT18	F3 disabled is unknown blockers, not no blockers	Nonempty safe coherent tasks; F4 enabled; F3 off or dependency view unavailable, not a verified empty set. Base progress/status/assignment/overdue facts work; blockers/reasons stay unknown with notice. F4 returns UNAVAILABLE/DEPENDENCY_INPUT_UNAVAILABLE, never an independent plan from edges=null. Valid current edges=[] is known no prerequisites; off/empty F4 gates are IT09/IT08.	FR-09/10/13; NFR-05; DR-04; D-01 pending
IT19	Member departure retains source work	One commit/revision removes active access and sets affected assigneeId=null; tasks, links, statuses, durations and notes remain. Historical note author reference remains valid; no automatic reassignment or task cancellation. Sole-leader policy remains open.	FR-02/06; NFR-04/05; DR-05
IT20	Task deletion does not silently destroy references	REFERENCE_POLICY_UNRESOLVED and no deletion under current conservative proposal. Later agreed cleanup must atomically involve F3/notes and revalidate graph. Do not add CANCELLED status or silently exclude the task from the denominator.	FR-03/06/08/12; OQ-05
IT21	Planned-start change remaps absolute finish	Conditional on the selected D-04 calendar/ceil(T) candidate: Duration/critical set unchanged; estimatedProjectFinish shifts by2 days and new result sourceRevision matches current snapshot. Old absolute finish is not simply relabeled current; deadline comparison refreshes.	FR-01/12/15; DR-02; D-04
IT22	Read-only current date refreshes basic overdue work	Conditional on the selected D-04 calendar/ceil(T) candidate: Equal date is not overdue under candidate rule; later date appears in FR-13 overdue list. No source revision increment, no CPM recomputation solely for clock progression, no dependency on F6 enablement. Missing taskDeadline is not evaluated rather than asserted on time.	FR-13; FR-15 scope; D-04
IT23	Matching viewer membership and assignment IDs	Task appears for M1; do not compare a global user ID directly with project membership ID. Unassigned tasks remain in project completion totals and disappear only from this personal-assignment list.	FR-02/06/13; NFR-04
IT24	Reliability trace and recovery remain honest	NFR-05 maps to Reliability, retained input/source and recovery, not maintainability/testability/explainability. Browser/device, performance and usability acceptance conditions remain unmeasured/open. No executed-test or approved-SLA claim.	NFR-01/02/03/04/05; OQ-07

13 Cross module validation matrix continued 25 to 27

IT25-IT27 below are additional NOT RUN design plans. The full matrix retains all24 earlier cases and gives recomputable precision/diagnostic inputs; conditional date clauses do not approve D-04.

ID	Scenario	Expected design outcome	Trace
IT25	End-to-end valid schedule and Blocked-critical warning	P25/current revision7; A2/B3/C6/D2 diamond. RelativeT10, criticalA/C/D, Bfloat3; raw2/1/1, completion50%, separate BlockedD with direct incompleteC only. Enabled F6 emits BLOCKED_CRITICAL_TASK forD. All sourceRevision values match7; no invented absolute finish or deadline conflict.	FR-09/10/11/13/14/15; DR-01/03/04
IT26	Duration precision and criticality	Conditional proposed NumericPolicy: integer/decimal plans, exact tied paths with near-zero binary artifacts and genuinely positive tiny float. Canonical recheck distinguishes true zero from display0.00. Invalid values cannot publish partial current facts; concrete limits/backend/tolerance validation remain D-03/D-07 PENDING. Full inputs/oracles are in IT26.	FR-10/11/12; DR-01/03; D-03/D-07
IT27	Invalid duration diagnostics and safe degradation	After feature/empty/current-dependency gates, invalid complete task durations give UNAVAILABLE/INVALID_INPUT, null schedule facts and authorized taskId/durationDays issue diagnostics. F2 save acceptance stays separate; reliable F5 and current MON-M3 blockers remain. F6 schedule-dependent types are unevaluated; denied access gives no private diagnostics.	FR-03/06/09/10/11/13/14/15; NFR-04/05; DR-04; D-03

14 Change evidence and next stage

Five independent ROUND2 attachments: Cross_Module_Interface_Contract, Requirement_Traceability_and_Priority_Matrix, Integration_Conflict_and_Decision_Log, Integration_Test_Matrix and Change_Log_and_Evidence_Index. These preserve source-page evidence, before/after mappings, canonical names, unresolved decisions and actual static/render checks. Targeted consistency repair 26 October 2026 updates scoped reasons, FR-13 mapping, dependency-view proof and IT10/IT11/IT18; previous evidence remains a dated historical snapshot. Weekly integration now adds IT25-27, centralized precision, authorized duration diagnostics, conditional ceil(T) dates and ten pending recommendations; prior evidence remains historical.

Original reports, diagrams,22 October review records and original JSON fixtures remain archived without altering their historical claims. Current navigation marks the two teammate source PDFs received and their editorial revisions integrated for review. Reconstructed DOCX sources from those PDFs are new editorial artifacts, not claims that the teammates authored or signed the corrections.

ROUND2 retains the existing architecture and valid Chapter6 discussion. It changes contradictory ownership, priority/NFR mapping, counts, availability/error behavior, version/date/identity fields, facade/module identifiers, lifecycle safeguards and inconsistent wireframe arithmetic. Detailed issue records distinguish URS-backed repair from proposed technical protocol and unresolved team/customer decisions.

Next: each owner reviews the editorial changes and canonical contract; the team/customer records D-01 scope and other open decisions; detailed interfaces/data/date policies and test subcases are then finalized. Implementation and executed integration/recovery/browser/performance checks belong to a later stage. No commit, push, publication, application test or approval occurred in this round.